

Teaching Monocular Depth the Metre with the Phone’s Own LiDAR: Fine-Tuned vs. Pretrained Depth Anything V2 for Room Reconstruction, Measured Against a Laser Scanner

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Abstract—Most phones have no LiDAR and must scan rooms with monocular depth, yet off-the-shelf “metric” models such as Depth Anything V2 (DA-v2) Metric-Indoor [1], trained on synthetic images, misplace a real room by 54.3 cm. We ask whether fine-tuning such a model with the phone’s own LiDAR depth, its weights learned from LiDAR alone (laser-scanner depth only chooses the checkpoint, on three rooms), makes it more accurate than its pretrained self, measured against the Faro Focus S70 laser scanner [2] that provides the ground truth of ARKitScenes [3]. We fine-tune only the DPT head [4] (the depth decoder; the image encoder stays frozen) on 24 rooms with iPad LiDAR labels, giving FT-LiDAR-24, and compare it with the pretrained model in one pipeline—per-frame depth fused into a truncated signed distance function (TSDF) volume [5], then meshed—where the depth source is the only variable, on rooms never used for training: six with a laser-scanned reference mesh and 20 more evaluated per frame. FT-LiDAR-24 lowers the Chamfer distance [6] (mean nearest-point distance between the reconstructed and reference meshes, both ways) from 54.3 cm to 15.7 cm (3.5 $\times$, better in every room) and AbsRel [7] (mean relative depth error) from 0.410 to 0.130 (better in 19/20 rooms), with no calibration or sensor at run time; a laser teacher is not measurably better. Its remaining error is a scale that swings from view to view, which fine-tuning does not remove, so it still trails iPad LiDAR (3.0 cm) and falls short of overview accuracy ($\geq 90\%$ of the room within 10 cm).

Index Terms—monocular depth estimation, transfer learning, metric depth, LiDAR supervision, TSDF fusion, 3D room reconstruction, ARKitScenes

I. INTRODUCTION

Scanning a room into a 3D model is already a consumer feature on iPhone Pro and iPad Pro (e.g., Apple RoomPlan [8]), and it relies on the *LiDAR sensor* for depth in metres. Devices with LiDAR are, however, a minority: non-Pro iPhones and almost every Android phone have none. Their alternative is *monocular depth estimation*—depth predicted from a single colour image—which has advanced rapidly from MiDaS [9] to Depth Anything [1], [10]. A product team must therefore know **by how many centimetres the room is wrong if LiDAR is replaced by a monocular model, and which use cases**

that still serves. Standard per-image benchmarks [7], which average the depth error over the pixels of single images, do not answer this, and end-to-end reconstruction systems [11]–[13] cannot, because they design the whole pipeline around their own model.

Some recent models output metres directly [1], [14], [15], but they are trained on images that do not come from a phone camera—DA-v2 Metric-Indoor on rendered rooms of the Hypersim dataset [16]—and in our experiments a room reconstructed with DA-v2 Metric-Indoor is 54.3 cm wrong on average. The direct remedy is to *fine-tune* the model on depth from the target device, which raises the question of where these labels come from: a survey-grade laser scanner [2], which few app makers own, or the *LiDAR already inside Pro devices*, whose depth an app can collect during normal use. If the latter works, devices with LiDAR can teach those without it.

We therefore ask whether DA-v2 Metric-Indoor fine-tuned with iPad LiDAR depth, its weights learned from LiDAR alone (laser-scanner depth only chooses the checkpoint, on three rooms that are not test rooms), is more accurate than the same model pretrained, when both are measured against the Faro laser scanner that provides the ground truth of ARKitScenes [3] on rooms never used for training. The study has four objectives: (1) to fine-tune the model on iPad LiDAR depth from 24 rooms, giving FT-LiDAR-24, and compare it with the pretrained model on held-out rooms, counting it as better if its mean error is lower and the per-room difference is significant at the 5% level under an exact Wilcoxon signed-rank test [17]; (2) to measure whether the source of the labels matters, by comparing Faro with LiDAR labels on the same 10 rooms (FT-Faro-10 and FT-LiDAR-10), 10 with 24 rooms of LiDAR labels, and a model that uses the camera’s focal length instead of in-domain labels (Depth Pro [15]); (3) to place FT-LiDAR-24 among the depth sources available at run time, namely iPad LiDAR and a relative model scaled per frame on sparse points or once per room; and (4) to find

the error that fine-tuning leaves, by fitting a depth scale to every frame and testing whether it follows time, which would indicate drift, or what the view shows.

Every comparison is scored in two ways. As a room mesh, it is scored by the Chamfer distance [6], the mean distance from each surface to the nearest point of the other, and by the F-score at 5 cm (F@5 cm) [18], which combines the share of the reconstruction lying within 5 cm of the reference with the share of the reference reconstructed within 5 cm. Per frame, it is scored by AbsRel [7], the mean relative depth error, and by the share of pixels whose predicted depth is within a factor of 1.25 of the true depth, which we call within-1.25 $\times$ [7]. For objective (3), the share of the reference reconstructed within 2, 5 and 10 cm stands for the accuracy needed for renovation measurement, furniture placement and a room overview, respectively, and a use case counts as served when that share reaches 90 %.

The scope is as follows. The data are home rooms of ARKitScenes captured with one iPad Pro model (colour images of 640×480 pixels, LiDAR depth of 256×192); the ground truth is the Faro depth distributed with the dataset, and iPad LiDAR is never used as ground truth. Only the DPT head [4], the depth decoder of DA-v2 Metric-Indoor Large, is fine-tuned, with one training recipe, while its image encoder stays frozen. Six test rooms are evaluated as meshes and per frame, and 20 further rooms per frame only, all disjoint from the rooms used for training and checkpoint selection. Every depth source passes through the same offline pipeline—per-frame depth, scale alignment, fusion into a truncated signed distance function (TSDF) volume [5], meshing—so the depth source is the only variable, and because every source and the reference share the camera poses recorded by ARKit, pose error is excluded. Learned reconstruction systems [11]–[13], other phone cameras including Android, real-time inference on the device and pose estimation are outside the scope.

Each objective has a matching expected benefit. The first tells an app maker whether data it can already collect from Pro-device users improves room scans on devices without LiDAR, with no calibration or extra sensor at run time; the second, whether labelling needs a laser scanner at all; the third, which product features a device without LiDAR can offer, together with a pipeline in which a new depth source is evaluated by changing one configuration entry; and the fourth, where to invest next—per-frame scale, training data or model size.

II. BACKGROUND AND RELATED WORK

This section explains the theory the study rests on—why one image does not fix depth in metres, how the model we fine-tune is built and trained, and how depth maps become a room—and places the study among prior work on each point.

A. Depth from a single image

In the pinhole model of a camera [19], the size of an object in the image is proportional to its real size divided by its distance, multiplied by the focal length of the lens. A room twice as large seen from twice as far therefore gives the same

image, so geometry alone cannot recover depth in metres from one image; a network can only infer it from learned cues, such as the usual height of a door or a table, and even these cues depend on the focal length, because the same object at the same distance looks larger through a longer lens. Eigen et al. [7] trained an early deep network for this task and, because of this ambiguity, measured its error on the logarithm of depth in a way that separates an error in the overall scale of the scene from an error in its shape. A weighted form of this scale-invariant logarithmic loss, known as SiLog, also trains ZoeDepth [14] and the fine-tunes of this study.

Two families of models handle the missing scale differently. Relative models give it up: MiDaS [9] trained one network on a mixture of datasets whose depths came from different sensors and units, which is possible only if the loss ignores the scale and offset of each image. Its output is therefore inverse depth (disparity) that is correct only up to an unknown scale and offset per image, and these two numbers must be found from other data before the output gives metres. Depth Anything [10] and Depth Anything V2 [1] (DA-v2) keep this output but train on far more images. Metric models output metres directly by learning the scale from labelled data. ZoeDepth [14] attaches metric heads, fine-tuned on NYU Depth v2 [20] and KITTI [21], to a relative model, and DA-v2 releases metric variants fine-tuned from its relative model, the indoor one (DA-v2 Metric-Indoor) on the rendered rooms of Hypersim [16]. A further line addresses the focal length: Metric3D [22] maps every image to a canonical camera with a fixed focal length, UniDepth [23] predicts the camera together with the depth, and Depth Pro [15] estimates the focal length when it is not given. These works report per-image metrics on benchmarks such as NYU Depth v2 and KITTI. Because such metrics average over independent images, they cannot show whether a model’s scale stays the same across the frames of one scan, which is what matters once the frames are fused into one room.

B. Architecture and training of Depth Anything V2

DA-v2 consists of an image encoder and a depth decoder. The encoder is DINOv2 [24], a vision transformer (ViT) [25]: it cuts the image into small square patches and lets every patch draw information from every other patch through a mechanism called self-attention. DINOv2 was pretrained without labels on a large curated collection of images, so that its features serve many tasks. The decoder is the Dense Prediction Transformer (DPT) [4], which takes features from several encoder layers, reassembles them into image-like maps at several resolutions and fuses these into one dense depth map. The released code splits the decoder into a neck, which reassembles and fuses, and a head, which outputs depth; as in the Introduction, “DPT head” in this paper means both.

DA-v2 is itself trained with a teacher. A large model is first trained on synthetic images only, whose depth is exact; it then labels about 62 million real images that have no depth, and the released models learn from these model-made labels, called pseudo-labels [1]. The DINOv2 authors report that depth can

be predicted well with the encoder kept frozen and only a DPT decoder trained on top [24]. Adapting a model trained on one task or domain to another in this way is called transfer learning [26]; training only the decoder, as we do, needs much less memory and keeps the general visual features that the encoder learned from millions of images.

C. Training on sensor depth

Supervised depth models have long been trained on depth measured by a sensor rather than drawn by people: NYU Depth v2 provides indoor depth from a Microsoft Kinect camera [20], KITTI provides street depth from a laser scanner mounted on a car [21], and Eigen et al. [7] trained on both. What differs between such studies is the quality of the sensor. On the iPhone Pro and iPad Pro, ARKit combines the LiDAR measurements with the colour image into a depth map of 256×192 pixels with a confidence value for every pixel [3]. Studies that tested this LiDAR against established surveying methods found it accurate to about 1 cm on objects larger than 10 cm [27] and adequate for rapid architectural mapping at a scale of 1:200 [28], which makes it a plausible teacher. ARKitScenes [3] provides this depth together with the depth of a Faro survey laser scanner [2] for the same rooms, registered to the same camera frames, and uses the Faro depth as ground truth for a task in which the LiDAR depth is upsampled with the help of the colour image. Prompt Depth Anything [29], the closest work to ours in data, likewise feeds iPhone LiDAR depth into a Depth Anything model as an extra input. In both, the LiDAR must be present whenever the model runs; in this study it is used only at training time, so the deployed model needs nothing but a colour image. Whether this noisier teacher helps a large pretrained model as much as a laser scanner does, when both are judged against an independent laser reference, is the subject of objectives (1) and (2).

D. From depth maps to a room

Fusing frames requires the camera pose of each frame, that is, its position and orientation. ARKit tracks the pose by visual-inertial odometry (VIO), which follows image features from frame to frame and combines them with the device’s motion sensor [30]. Because the motion sensor measures acceleration in metric units, the tracked feature points have positions in metres, and such sparse points can supply, frame by frame, the scale and offset that a relative model lacks. Combining learned depth with the sparse geometry of a tracking system goes back to CNN-SLAM [31], and MonoSDF [32] fits a scale and an offset to every image before using monocular depth to guide surface reconstruction; the sparse-point alignment of Section III-C follows this line.

The room itself is built by fusion into a truncated signed distance function (TSDF) [5]. Space is divided into small cubes called voxels; each voxel stores its distance to the nearest observed surface, positive in front of the surface and negative behind it, cut off (truncated) beyond a few voxels, and this value is averaged over every frame that sees the voxel. The surface is then extracted where the average crosses zero, with

the marching cubes algorithm [33]. KinectFusion [34] made this fusion run in real time on Kinect depth, and we use the implementation in Open3D [35]. Averaging suppresses noise that is random from frame to frame, but frames that disagree in scale place the same wall at different distances, and the average then blurs or thickens it; the consistency of scale across frames, which per-image metrics do not measure, therefore shapes the room. TSDF fusion also does not depend on where the depth in metres comes from, which makes it a fair fixed stage for every depth source.

Learned reconstruction systems replace parts of this pipeline with networks. Atlas [11] predicts the TSDF directly from features of many posed images, NeuralRecon [12] does so incrementally and in real time, and SimpleRecon [13] estimates depth from several neighbouring frames before classic fusion; all are evaluated on ScanNet [36], a large collection of indoor colour-and-depth scans. Each needs several frames per prediction and a network trained for its own pipeline, so they answer a different question from ours—what room a single-image model yields when nothing else in the pipeline changes—and we do not compare against them.

E. Measuring a reconstructed room

A reconstructed room is compared with a reference surface through distances between points sampled on both. The Chamfer distance [6] averages, in both directions, the distance from each point to the nearest point of the other surface. Tanks and Temples [18] instead fixes a distance threshold and reports precision, the share of the reconstruction within the threshold of the reference; recall, the share of the reference within the threshold of the reconstruction; and their harmonic mean, the F-score. A threshold ties the score to a use, which is how the Introduction links 2, 5 and 10 cm to renovation measurement, furniture placement and a room overview. Per frame, we use AbsRel and within- $1.25\times$ from the single-image literature [7].

F. Position of this study

In summary, prior work reports the per-image accuracy of depth models on standard benchmarks, adapts metric models with labels from synthetic images or from the sensors of public datasets, feeds mobile LiDAR to a model at run time, or tests mobile LiDAR itself against surveying methods. We did not find a study that combines these threads as this one does: it uses the LiDAR of a consumer device only as a training teacher, compares it with a survey laser scanner as teacher on the same rooms, and judges both against an independent laser reference, in centimetres on fused rooms as well as per frame.

III. METHODOLOGY

The study compares depth sources inside one fixed reconstruction pipeline (Fig. 1). For every room and every depth source it (1) reads the colour frames and camera poses, (2) obtains a depth map for every frame from the source, (3) converts that depth to metres if the source does not give metres, (4) fuses the frames into a room mesh, and (5) measures the per-frame depth and the mesh against the Faro

TABLE I
TEST ROOMS, EVALUATED IN 3D AND PER FRAME. FRAMES: FARO
FRAMES ON WHICH EVERY ROW IS EVALUATED.

Scan ID	Room	Extent (m)	Frames
42444474	kitchen/dining with glass (development room)	$9.9 \times 5.1 \times 4.2$	236
47333774	standard bedroom	$5.1 \times 4.9 \times 3.3$	375
47115299	large living room	$8.1 \times 6.1 \times 2.5$	324
42897743	bathroom (mirror, tiles)	$8.0 \times 3.4 \times 3.6$	448
47429736	bedroom with closets	$6.6 \times 6.4 \times 2.7$	504
45261556	kitchen (glossy cabinets)	$7.1 \times 6.1 \times 2.4$	465

reference. Only steps (2) and (3) change between the rows of a result table; each is chosen by one entry of a configuration file, and every later step runs the same code with the same settings. The fine-tuned models enter at step (2) as ordinary depth sources. This section describes the data, each step, the fine-tuning, the metrics and statistics, and the experiments.

A. Data, rooms and reference

ARKitScenes [3] provides, for the same room, the colour video of an iPad Pro (640×480 pixels), the depth of its LiDAR as produced by ARKit (256×192 pixels, with a confidence of low, medium or high for every pixel), the camera pose of every frame from ARKit’s visual-inertial odometry, and depth from a Faro laser scanner [2] registered to those poses (1920×1440 pixels, about four frames per second). **The ground truth is this Faro depth as distributed.** For per-frame evaluation it is resampled to 256×192 by taking the nearest pixel, so every reference value is a measured one. For 3D evaluation, since no laser mesh is distributed, we fuse the Faro depth of every frame at 1 cm voxels, once per room, into a reference mesh with the fusion of Section III-D. Because this reference and every depth source use the same poses, pose error is excluded and every difference between rows is due to the depth source. iPad LiDAR is never used as ground truth.

The six test rooms (Table I) span a standard bedroom, large rooms and rooms with mirrors or glossy surfaces. Room 42444474 was used while the pipeline was developed, so its numbers are not held out; the per-room results let the reader check every conclusion on the other five. The rooms for fine-tuning were drawn at random (seed 0) from ARKitScenes scans of 60 to 120 s, excluding every visit of a test room, a visit being the capture session of one home, whose videos show the same rooms: 10 training rooms with both Faro and LiDAR depth, 14 more with LiDAR only, and 3 rooms of the dataset’s Validation fold for choosing the checkpoint. Exp. 7 adds 20 further Validation-fold rooms, one video per visit, drawn with seed 0 and excluding every visit of the training, checkpoint-selection and test rooms. They decided nothing and are evaluated per frame only, on every second Faro frame (2,090 frames in total).

B. Depth sources

Four kinds of depth source are compared. (1) The Faro depth itself shows what the pipeline loses when depth is perfect: at 4 cm voxels it still differs from the 1 cm reference, and that difference is the pipeline’s own discretisation loss.

TABLE II
SCALE ALIGNMENTS. EVERY ALIGNMENT OTHER THAN “NONE” FITS ONE
SCALE AND ONE OFFSET IN INVERSE DEPTH BY TRIMMED LEAST
SQUARES.

Alignment	Fitted on, and what the row shows
none	nothing; used for Faro, LiDAR and metric models
per-frame oracle	the frame’s own Faro depth; the best the model’s shape allows (not usable in a product)
once per room	Faro depth of 10 frames spread evenly over the scan, fitted once and kept for every frame; an upper bound of a one-off calibration
sparse points	200 metric points per frame, standing for the feature points of VIO (Section II); ARKitScenes stores none, so they are sampled from LiDAR pixels of high confidence, which are cleaner than triangulated features; an upper bound of a VIO-based system

(2) The iPad LiDAR depth as recorded by ARKit is what a Pro device uses today. (3) Relative monocular models need the scale alignment of Section III-C: DA-v2 Large in the main experiments, and DA-v2 Small and MiDaS small [9] when model size is varied. (4) Metric monocular models are used with no alignment: DA-v2 Metric-Indoor Large, pretrained; its three fine-tunes (Section III-E); and Depth Pro [15], given the camera’s true focal length of 533 pixels, its most favourable setting. Its implementation was checked on a synthetic room with exact depth; on five frames of room 47115299 it was also run with its own estimate of the field of view and scored after removing the scale of each frame. ARKitScenes stores the frames of an iPad held upright rotated by a quarter turn, so every monocular model receives the frame rotated upright and its output is rotated back. Every source is resampled by nearest pixel to the LiDAR resolution of 256×192 , so that all sources are fused and scored on the same grid.

C. Scale alignment

A relative model gives inverse depth that is known only up to a scale and an offset per image (Section II). Alignment finds the scale and offset that make the model’s inverse depth best match a set of reference depths, also inverted, and converts the result back to depth. Fitting in inverse depth rather than in depth follows how these models are trained; on the development room it reduced the Chamfer distance of the per-frame oracle from 15.2 to 5.4 cm. The fit is by least squares, repeated three times, each time leaving out the 20 % of pixels with the largest residuals, because depth edges produce a few very large residuals that would otherwise pull the fit. The four alignments differ only in the reference depths they fit against (Table II); a metric source is always used with no alignment, and the pipeline refuses to run otherwise.

For sparse points the fit is repeated twice and leaves out the worst 10 % of points, and a frame with fewer than 20 usable points reuses the previous frame’s scale and offset, as a deployed system would when tracking briefly fails.

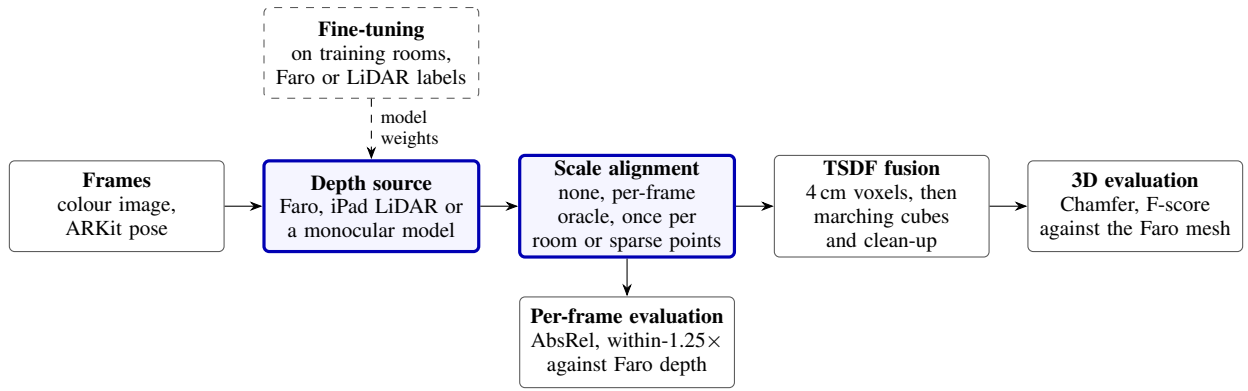


Fig. 1. Pipeline. The two highlighted steps are the only ones that change between the rows of a result table, each by one configuration entry; every step after them is fixed. Fine-tuning (dashed) happens once, beforehand, and only supplies the weights of a monocular model.

D. Fusion into a room mesh

The depth maps are fused by TSDF fusion (Section II) in Open3D [35] with 4 cm voxels, a truncation distance of 12 cm (three voxels) and depths from 0.1 to 5 m. For every source, only the pixels where Faro has depth are integrated, so no source is penalised for seeing parts of the room the reference does not cover, and every pixel has the same weight (only Exp. 5 weights pixels by LiDAR confidence). The mesh is extracted with marching cubes [33], and connected pieces of fewer than 1000 triangles are removed as floating noise. Every row uses the ARKit poses of the dataset.

E. Fine-tuning

Three fine-tunes of DA-v2 Metric-Indoor Large share one recipe and differ only in their labels. FT-Faro-10 learns from Faro depth on the 10 training rooms (3594 frames), FT-LiDAR-10 from iPad LiDAR depth on the same 10 rooms (8656 frames), and FT-LiDAR-24, the main model, from LiDAR depth on all 24 rooms (19 896 frames). LiDAR pixels of low confidence are dropped, and label depths beyond 10 m are ignored. The two teachers do not see the same frames: Faro depth exists at about four frames per second, whereas LiDAR depth comes with every colour frame, of which we take every third (about ten per second), so FT-LiDAR-10 sees about 2.4 times as many distinct views as FT-Faro-10.

The image encoder is frozen and only the DPT decoder, neck and head, is trained (Section II). Each training image is rotated upright, resized so that its short side is 518 pixels and cropped to 518×518 at a random position; it is never zoomed, because enlarging an image changes the apparent size of objects and would teach wrong metres. Horizontal flips (probability 0.5) and colour jitter (strength 0.2) add variety. The loss is SiLog (Section II): for each image, the variance of the difference between the logarithms of predicted and label depth, which scores shape, plus 0.15 times the square of its mean, which scores the overall scale; the square root of this sum is multiplied by 10. The optimiser is AdamW [37], a form of the Adam optimiser that applies weight decay separately from the gradient step, with a learning rate of 5×10^{-5} , weight

decay 0.01, a linear warm-up over 200 steps and then a cosine decay to zero [38]. Each step uses 8 images (2 at a time, with gradients accumulated over 4), and gradients are clipped at a norm of 1. Training runs for 6000 steps for FT-Faro-10 and FT-LiDAR-10 and 12 000 for FT-LiDAR-24. Every 500 steps the model is scored by AbsRel against the Faro depth of every fifth frame of the three checkpoint-selection rooms, and the checkpoint with the lowest AbsRel is kept; the weights of the LiDAR-taught models are therefore learned from LiDAR alone, but their checkpoint is chosen with Faro depth on these three rooms. A 6000-step run takes about an hour on an NVIDIA RTX 3070 Ti GPU. At test time a fine-tuned model is used exactly like the pretrained one: colour image in, metres out, no alignment and the same input preparation.

F. Metrics and statistics

3D. 200 000 points are sampled uniformly on each mesh. Accuracy is the mean distance from a point of the reconstruction to the nearest point of the reference, completeness the mean distance in the other direction, and the Chamfer distance [6] their mean. Precision, recall and F-score [18] (Section II) are computed at thresholds of 2, 5 and 10 cm, with F@5 cm as the headline. For objective (3), a use case counts as served when recall at its threshold is at least 0.9, as partly served from 0.75 to 0.9, and as not served below 0.75. **Per frame.** Each depth map is compared with the Faro depth on the 256×192 grid, over the pixels where Faro depth lies between 0.1 and 5 m and the source gave a value; a prediction outside that range is clipped to it rather than dropped, so a model that places a wall too far away is still scored on that wall. We report AbsRel, within-1.25 $\times$ and the root mean square error (RMSE) of depth [7], averaged over the frames of a room and then over rooms. **Statistics.** Tables give the mean $\pm$ standard deviation over rooms. Because the rooms, not the frames, are the independent units, two models are compared by their per-room values with the exact two-sided Wilcoxon signed-rank test [17]. The test ranks the per-room differences by size and asks whether the ranks of positive and negative differences are balanced; “exact” means the p-value is computed from every possible assignment of signs rather than

TABLE III
EXPERIMENTS. IN EACH, THE LISTED VARIABLE CHANGES AND EVERYTHING ELSE IS FIXED.

Exp.	Obj.	Variable; rooms and evaluation
6	1, 2	pretrained vs. FT-LiDAR-24; FT-Faro-10, FT-LiDAR-10 and Depth Pro; 6 test rooms, 3D and per frame
7	1	pretrained vs. FT-LiDAR-24, iPad LiDAR for reference; 20 further rooms, per frame
1	3	depth source and alignment: Faro, LiDAR, DA-v2 Large with each alignment, pretrained DA-v2 Metric; 6 test rooms, 3D and per frame
–	4	per-frame scale analysis (Section III-H); 6 test rooms, every fifth frame
2–5	–	voxel size; frame stride; model size; LiDAR confidence; 6 test rooms, 3D

from an approximation, which matters for few rooms. With six rooms the smallest attainable p-value is $2/64 \approx 0.031$, so the difference is significant at the 5 % level only if one model wins in all six.

G. Experiments

Table III lists the experiments and the objective each serves. Exp. 6 and 7 answer the main question (objectives 1 and 2), Exp. 1 places the models among the run-time depth sources (objective 3), and Exp. 2–5 test how sensitive the pipeline is to its own settings. Times in Exp. 4 were measured with unoptimised PyTorch on an Apple Silicon laptop and are used only as ratios.

H. Per-frame scale analysis

Objective (4) asks which error fine-tuning leaves and whether it comes from time or from the view. On every fifth frame of the six test rooms (471 frames) we compute, for the pretrained metric model and for FT-LiDAR-24, the ratio of Faro depth to predicted depth, taken as the median over the frame’s pixels; its median over the frames of a room shows how far the model’s scale is off, and the ratio of its 95th to its 5th percentile over those frames shows how much the needed scale varies within one room. We then compute AbsRel with increasing help: none; one scale and offset per room, fitted on 10 frames as in Table II; the right scale for every frame (the frame’s median ratio, no offset); and the per-frame oracle scale and offset. For the relative model DA-v2 Large we fit the per-frame oracle scale on the same frames and compute its Pearson correlation coefficient [39], a measure of linear association from -1 to $+1$, with the frame’s time in the scan and with the median Faro depth of the frame. Drift, an error that builds up over the scan, would show as a correlation with time of the same sign in every room; dependence on the view would show as a consistent correlation with the depth of the scene in front of the camera.

IV. RESULTS

This section reports the measurements of the experiments in Table III, grouped by objective; what they mean is discussed in

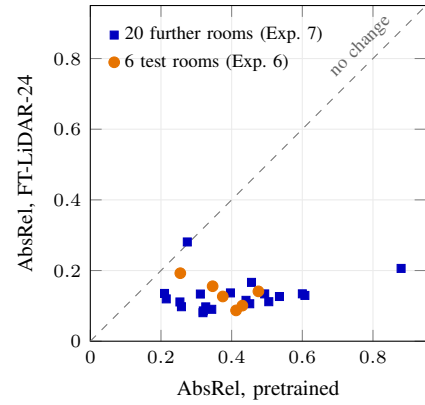


Fig. 2. Per-room AbsRel against Faro: pretrained DA-v2 Metric-Indoor (x) and FT-LiDAR-24 (y). Points below the dashed diagonal are rooms where FT-LiDAR-24 has the lower AbsRel.

Section V. Distances are in centimetres, values are means over rooms, and $\pm$ gives the standard deviation over rooms. When two models are compared, we state in how many rooms one is better and give the p-value of the exact Wilcoxon signed-rank test on the per-room values (Section III-F).

A. Fine-tuned against pretrained (objective 1)

On the six test rooms (Table IV), FT-LiDAR-24 lowered the Chamfer distance from 54.3 to 15.7 cm. It was lower in all six rooms, by 38.6 ± 8.6 cm ($p = 0.031$, the smallest p-value attainable with six rooms). Accuracy fell from 70.3 to 17.3 cm and completeness from 38.3 to 14.0 cm, and F@5 cm rose from 0.03 to 0.24, higher in all six rooms ($p = 0.031$). Per frame, AbsRel fell from 0.382 to 0.134, lower in all six rooms ($p = 0.031$, Table VII), and within- $1.25\times$ rose from 0.277 to 0.812, higher in five rooms ($p = 0.063$); in room 47115299 it fell from 0.74 to 0.60.

On the 20 further rooms of Exp. 7 (Table V, Fig. 2), AbsRel fell from 0.410 to 0.130, lower in 19 of the 20 rooms ($p = 3.8 \times 10^{-6}$), and within- $1.25\times$ rose from 0.333 to 0.827, higher in 19 rooms ($p = 9.5 \times 10^{-6}$). In those 19 rooms, the AbsRel of FT-LiDAR-24 was 21 to 64 % of the pretrained model’s. In the remaining room, 48018730, whose pretrained AbsRel was the fifth lowest of the 20, AbsRel went from 0.275 to 0.281 and within- $1.25\times$ from 0.53 to 0.35. For comparison, iPad LiDAR has AbsRel 0.019 on these rooms and a Chamfer distance of 3.0 cm on the test rooms, and DA-v2 Large aligned on sparse points has 6.0 cm, 9.7 cm less than FT-LiDAR-24 (Table IV).

B. Source of the labels (objective 2)

The three fine-tunes reached Chamfer distances of 14.6 cm with Faro labels (FT-Faro-10), 16.6 cm with LiDAR labels on the same rooms (FT-LiDAR-10) and 15.7 cm with LiDAR labels on 24 rooms (FT-LiDAR-24), and AbsRel of 0.136, 0.136 and 0.134 (Table IV). Per room (Table VII), FT-Faro-10 had a lower Chamfer distance than FT-LiDAR-10 in four rooms ($p = 0.69$) and FT-LiDAR-24 lower than FT-LiDAR-10 in five ($p = 0.44$); FT-Faro-10 was lower than FT-LiDAR-24 in three rooms, higher in two and within 0.1 cm in one ($p = 0.56$).

TABLE IV

EXP. 6 ON THE SIX TEST ROOMS: COLOUR IMAGES ONLY, NO SCALE ALIGNMENT, MEASURED AGAINST FARO. THE LAST TWO ROWS NEED A SENSOR AT RUN TIME AND ARE GIVEN FOR REFERENCE (EXP. 1). ACC.: ACCURACY; COMPL.: COMPLETENESS; F@5 AND R@10: F-SCORE AT 5 CM AND RECALL AT 10 CM.

Model	Labels for training	Chamfer	Acc.	Compl.	F@5	R@10	AbsRel	within-1.25×	RMSE
DA-v2 Metric-Indoor, pretrained	— (rendered rooms, Hypersim)	54.3 ± 10.6	70.3	38.3	0.03	0.10	0.382	0.277	49.3
Depth Pro, true focal length	—	155.4 ± 39.0	229.0	81.8	0.00	0.02	2.007	0.170	206.9
FT-Faro-10	Faro, 10 rooms	14.6 ± 4.0	13.2	15.9	0.31	0.57	0.136	0.780	20.7
FT-LiDAR-10	iPad LiDAR, 10 rooms	16.6 ± 4.7	18.8	14.3	0.23	0.49	0.136	0.848	19.8
FT-LiDAR-24	iPad LiDAR, 24 rooms	15.7 ± 4.0	17.3	14.0	0.24	0.51	0.134	0.812	20.2
ref.: iPad LiDAR (sensor)	—	3.0 ± 0.6	2.2	3.9	0.93	0.97	0.018	0.997	3.9
ref.: DA-v2 Large + sparse points	—	6.0 ± 1.7	6.4	5.5	0.74	0.91	0.036	0.985	7.4

TABLE V

EXP. 7: PER-FRAME METRICS ON 20 FURTHER ROOMS OF THE VALIDATION FOLD (2,090 FRAMES), USED FOR NOTHING ELSE; MEAN ± STD OVER ROOMS.

Model	AbsRel	within-1.25×	RMSE (cm)
DA-v2 Metric-Indoor, pretrained	0.410 ± 0.164	0.333 ± 0.216	50.9
FT-LiDAR-24	0.130 ± 0.046	0.827 ± 0.135	18.4
ref.: iPad LiDAR (sensor)	0.019 ± 0.002	0.997 ± 0.002	3.8

TABLE VI

EXP. 1, SIX TEST ROOMS: ONLY THE DEPTH SOURCE AND THE SCALE ALIGNMENT CHANGE (FT-LiDAR-24 FROM EXP. 6). RECALL AT 10, 5 AND 2 CM: ROOM OVERVIEW, FURNITURE PLACEMENT, RENOVATION MEASUREMENT; FOR ROWS USABLE AT RUN TIME, ✓ SERVED (≥ 0.9), ~ PARTLY (0.75–0.9), × NOT (< 0.75).

Source	Alignment	Chamfer	F@5	AbsRel	Recall at		
					10 cm	5 cm	2 cm
Faro	none	2.1 ± 0.5	0.97	0	0.98	0.96	0.84
iPad LiDAR	none	3.0 ± 0.6	0.93	0.018	✓ 0.97	✓ 0.91	× 0.52
DA-v2 Large	per-frame oracle	5.3 ± 1.3	0.79	0.032	0.93	0.82	0.47
DA-v2 Large	sparse points	6.0 ± 1.7	0.74	0.036	✓ 0.91	~ 0.77	× 0.39
DA-v2 Large	once per room	22.9 ± 2.1	0.23	0.237	× 0.47	× 0.27	× 0.09
DA-v2 Metric-Indoor	none	54.3 ± 10.6	0.03	0.382	× 0.10	× 0.04	× 0.01
FT-LiDAR-24	none	15.7 ± 4.0	0.24	0.134	× 0.51	× 0.28	× 0.09

For none of these three pairs was the difference in Chamfer distance, F@5 cm, AbsRel or within-1.25× significant (every $p \geq 0.09$). Fig. 3 shows the error of each surface point for every row and room.

Depth Pro, given the camera’s true focal length and no fine-tuning, reached 155.4 cm and AbsRel 2.007, both higher than the pretrained DA-v2 Metric-Indoor in all six rooms ($p = 0.031$). In the check on five frames of room 47115299 (Section III-B), its own estimate of the horizontal field of view was 48.0 to 48.7°, against 61.6° for the camera, and after removing the scale of each frame its AbsRel was 0.24 to 0.54, against 0.04 to 0.13 for pretrained DA-v2 Metric-Indoor on the same frames.

C. Place among the run-time depth sources (objective 3)

Through the same pipeline (Table VI), Faro depth itself gave 2.1 cm. iPad LiDAR gave 3.0 cm, 0.98 ± 0.25 cm above Faro per room and above it in every room; its recall at 2 cm was 0.52, against 0.84 for Faro. DA-v2 Large with the per-frame oracle gave 5.3 cm, 2.29 ± 1.11 cm above LiDAR per room; aligned on sparse points it gave 6.0 cm, and aligned

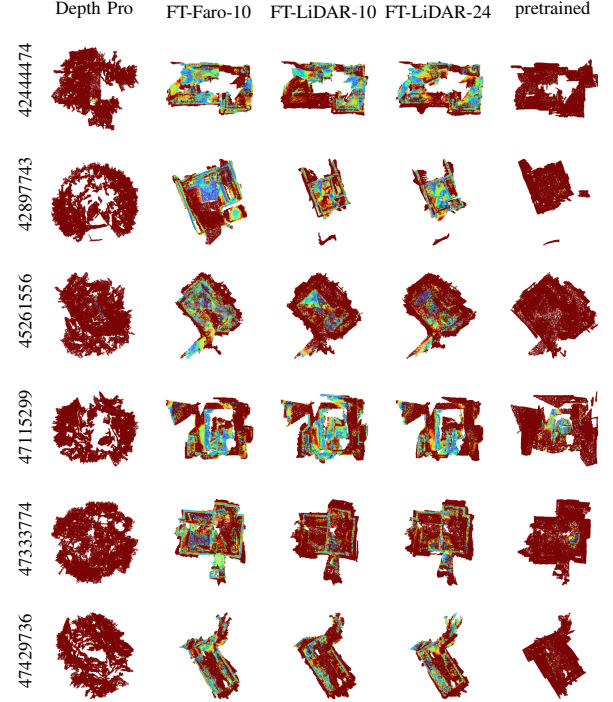


Fig. 3. Exp. 6 seen from above: every point of the reconstructed mesh coloured by its distance to the Faro mesh, from dark blue (0 cm) through green and yellow to dark red (10 cm or more). Rows are rooms, by scan ID; columns are models.

once per room 22.9 cm, between 19.9 and 26.0 cm in every room. The pretrained DA-v2 Metric-Indoor gave 54.3 cm; in room 45261556 it had its highest Chamfer distance (71.0 cm) while the per-frame oracle had its second lowest there (4.5 cm; Table VII). By the thresholds of Section III-F, iPad LiDAR served the overview and furniture placement, sparse points served the overview and partly served furniture placement, and no row, FT-LiDAR-24 included, served renovation measurement.

D. Per-frame scale (objective 4)

For the pretrained model, the median ratio of Faro to predicted depth was 0.69 to 0.85 in the six rooms, so its predicted depth was 1.18 to 1.45 times the Faro depth; for FT-LiDAR-24 the ratio was 0.93 to 1.21 (Table VIII). Within

TABLE VII
PER TEST ROOM, IN THE ORDER OF TABLE I (FIRST FOUR DIGITS OF THE SCAN ID). DA-v2 L: DA-v2 LARGE.

	4244	4733	4711	4289	4742	4526	mean
<i>Chamfer distance (cm)</i>							
Faro	1.8	1.2	1.9	2.8	2.4	2.3	2.1
iPad LiDAR	2.8	2.4	2.9	4.1	3.1	3.0	3.0
DA-v2 L, per-frame oracle	3.8	6.2	4.8	7.5	5.2	4.5	5.3
DA-v2 L, sparse points	4.5	6.1	5.5	9.1	5.9	4.6	6.0
DA-v2 L, once per room	24.1	26.0	19.9	23.2	21.4	23.1	22.9
pretrained DA-v2 Metric	48.5	61.6	41.5	48.9	54.1	71.0	54.3
Depth Pro	131.1	165.2	172.9	196.4	177.9	88.9	155.4
FT-Faro-10	10.8	13.8	17.6	21.1	12.4	11.9	14.6
FT-LiDAR-10	13.9	19.0	12.8	12.5	16.6	24.6	16.6
FT-LiDAR-24	12.8	16.7	17.3	12.2	12.4	22.5	15.7
<i>AbsRel</i>							
pretrained DA-v2 Metric	0.375	0.346	0.255	0.412	0.476	0.430	0.382
FT-LiDAR-24	0.127	0.156	0.193	0.087	0.141	0.101	0.134

TABLE VIII
PER-FRAME SCALE OF THE PRETRAINED MODEL AND FT-LiDAR-24, EVERY FIFTH FRAME OF THE SIX TEST ROOMS (471 FRAMES). RATIO: FARO DEPTH OVER PREDICTED DEPTH, MEDIAN OVER THE PIXELS OF A FRAME.

	Pretrained	FT-LiDAR-24
ratio, median over the frames of a room (range over rooms)	0.69–0.85	0.93–1.21
ratio, 95th over 5th percentile within a room (mean)	1.55	1.52
AbsRel, no help	0.381	0.134
AbsRel, one scale and offset per room from 10 frames	0.127	0.117
AbsRel, the right scale on every frame	0.045	0.051
AbsRel, per-frame oracle scale and offset	0.035	0.042

a room, the 95th percentile of the ratio was on average 1.55 times its 5th percentile for the pretrained model and 1.52 times for FT-LiDAR-24; in room 47115299 the pretrained model’s ratio ranged from 0.43 to 1.35. With no help, FT-LiDAR-24 had AbsRel 0.134, against 0.127 for the pretrained model with one scale and offset per room fitted on Faro depth. With the right scale on every frame, AbsRel fell to 0.045 for the pretrained model, 97 % of the way from no help to the oracle, and to 0.051 for FT-LiDAR-24.

For the relative model DA-v2 Large, the per-frame oracle scale varied within a room by a factor of 2.6 to 4.9 between its 5th and 95th percentiles. Its correlation with the frame’s time in the scan ranged from -0.40 to $+0.30$, positive in three rooms and negative in three; its correlation with the frame’s median Faro depth ranged from -0.75 to -0.27 , negative in all six (Fig. 4).

E. Sensitivity of the pipeline (Exp. 2–5)

Table IX lists the sensitivity experiments. DA-v2 Small gave a Chamfer distance 1.0 cm higher than DA-v2 Large and was 6.4 times faster per room; at four frames per second, processing took 3.6 times the length of the video with DA-v2 Large and 0.6 times with DA-v2 Small, of which fusion was less than 3 %.

V. DISCUSSION AND CONCLUSION

This section relates the results to prior work, interprets the error that remains, draws implications, states the limitations and future work, and concludes objective by objective.

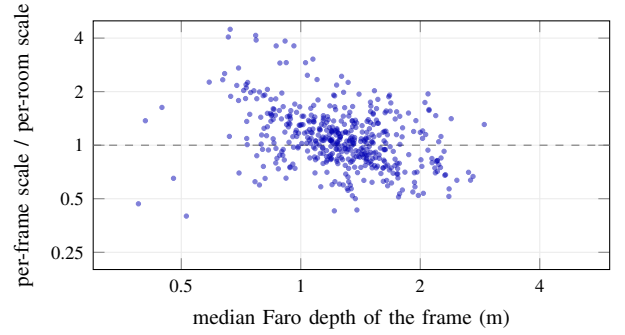


Fig. 4. Relative model DA-v2 Large, every fifth frame of the six test rooms: the per-frame oracle scale divided by the once-per-room scale, against the median Faro depth of the frame (469 frames; two frames whose fit gave no positive scale are left out). Pearson correlation of the logarithms: -0.37 .

TABLE IX
SENSITIVITY EXPERIMENTS ON THE SIX TEST ROOMS; ONE SETTING CHANGES, EVERYTHING ELSE AS IN SECTION III.

Exp.	Result
2 voxel size	Faro depth; 2 / 4 / 8 cm: Chamfer 1.2 / 2.1 / 2.8 cm; fusion at most 2 s per room at every size
3 frame stride	Faro depth; every 1st / 5th / 10th frame (4 / 0.8 / 0.4 per second): accuracy 1.3 / 1.2 / 1.2 cm, completeness 2.8 / 7.5 / 16.6 cm, F@5 cm 0.97 / 0.91 / 0.83
4 model size	per-frame oracle; DA-v2 Large (335 M parameters) / Small (25 M) / MiDaS small (21 M): Chamfer 5.33 / 6.35 / 13.24 cm; 348 / 55 / 16 s per room
5 LiDAR confidence	Chamfer 3.04 cm with every pixel, 3.07 cm without low-confidence pixels, 3.66 cm with high-confidence pixels only (completeness 3.85 $\rightarrow$ 5.23 cm), 3.04 and 3.07 cm with pixels weighted by confidence

A. Relation to prior work

DA-v2 Metric-Indoor learned metres from the rendered rooms of Hypersim [1], [16] and placed every real test room 1.18 to 1.45 times too far. ZoeDepth [14] and DA-v2 adapt their metric heads to a target domain by fine-tuning on labelled data; our result shows that the same remedy works with the target device’s own LiDAR as the labels. As the DINOv2 authors did for depth [24], we trained only the decoder on the frozen encoder, and the shape of the prediction barely changed (per-frame oracle AbsRel 0.035 against 0.042), which fits the view that the encoder’s features transfer [26] and the head mainly learned the scale.

Sensor depth has long served as a training label, from Kinect indoors to laser scanners on cars [7], [20], [21]. The roughly 1 cm that phone LiDAR adds to the Faro depth in our pipeline agrees with the centimetre-level accuracy reported against surveying methods [27], [28], and its error as a label (AbsRel 0.018) is an order of magnitude below the error left in the fine-tuned model (0.134), which may explain why a laser teacher gave no measurable gain. Prompt Depth Anything [29] and the upsampling task of ARKitScenes [3] use the LiDAR at run time; our model needs only a colour image, at a large price (15.7 cm against 3.0 cm).

Camera-aware models [15], [22], [23] treat the unknown focal length as the main obstacle to metric depth. For Depth Pro on these 640×480 indoor frames, the true focal length did not make it usable, and its error without scale stayed several times that of DA-v2 Metric-Indoor; we did not test Metric3D or UniDepth, so this does not extend to the approach in general. Per-image benchmarks cannot show whether the scale stays the same across the frames of a scan (Section II); we found that it does not, for relative and metric models alike. Video Depth Anything [40] targets such consistency across video frames, and sparse metric points, used as in CNN-SLAM and MonoSDF [31], [32], came within 0.6 cm of the per-frame oracle in their upper-bound form. Learned reconstruction systems [11]–[13] use several frames per prediction and other data, so our numbers are not comparable with theirs.

B. Why a scale error remains

Fine-tuning learned the domain’s multiplier: FT-LiDAR-24 without calibration (AbsRel 0.134) is on a par with the pretrained model given one scale and offset per room fitted on ground truth (0.127), so what a one-off calibration provides has moved into the weights of the head. The swing from frame to frame did not shrink (1.52 against 1.55). A likely reason follows from the pinhole model (Section II): views such as a close-up of a table top or a blank wall look alike at different distances and give no cue to absolute size, so the model falls back on a typical indoor distance; the negative correlation between the needed scale and the depth of the view, found in every room, fits this reading. If it is right, more or better labels will not remove the swing, and metric data at run time will be needed. TSDF fusion [5] then averages the same surface placed at several distances into a thicker, blurred surface, consistent with a moderate per-frame error becoming 15.7 cm in Chamfer distance.

C. Implications for practice

For an app on a device without LiDAR, the results suggest an order of work: per-frame metric scale first (within 0.6 cm of the per-frame oracle, against 22.9 cm with one calibration per room, an upper bound because the points came from LiDAR, not from VIO), then a fine-tuned model, which removes the domain bias with nothing at run time, and model size last (DA-v2 Small was 1.0 cm worse than Large under the per-frame oracle and 6.4 times faster). The labels can be collected from Pro devices during normal use, since ARKit delivers colour, LiDAR depth and pose together. At 3.6 times the video length on a laptop, the pipeline as tested runs offline.

D. Limitations

The evaluation is small: six test rooms in 3D, one used during development, and 26 rooms per frame, all homes captured with one iPad model; offices, empty rooms, low light and other phone cameras are absent, and the bathroom with a mirror was the hardest room for every source with the right scale (Table VII). With six rooms the Wilcoxon test detects only a difference present in every room, so the teacher

comparison cannot rule out differences of the size seen in the means, one to two centimetres. The main comparison was fixed after the six test rooms had been evaluated; the 20 rooms of Exp. 7 were added afterwards so that the conclusion also rests on rooms that decided nothing. The reference shares the ARKit poses of every row, so a real phone system would add pose drift, and completeness is relative to what Faro saw. Only the head of one model was fine-tuned, with one recipe, and FT-Faro-10 and FT-LiDAR-10 differ in the number of views as well as the labels. Checkpoints were chosen with Faro depth, also for the LiDAR-taught models, on three rooms; FT-LiDAR-24 became the main model because it was best there and its labels need no laser scanner, but that ranking did not reproduce on the test rooms, and its validation AbsRel changed by up to 0.1 between checkpoints 500 steps apart. The once-per-room and sparse-point alignments are upper bounds, calibrated on ground truth and on clean LiDAR samples. Depth Pro was run through one implementation, and its focal-length check covered five frames of one room. The thresholds that link recall to use cases are our illustration, not an industry standard, and times are unoptimised and meaningful only as ratios.

E. Future work

In order of expected impact, future work should (1) combine FT-LiDAR-24 with per-frame metric points from the device’s tracker, for which Section IV-D gives AbsRel 0.051 with the right scale on every frame; (2) choose checkpoints with LiDAR depth, so that no step needs a laser scanner, and compare teachers on identical frames and more test rooms; (3) train and evaluate on more phone cameras, including Android; (4) evaluate with real ARKit or ARCore feature points and a reference captured by the app itself; and (5) test fine-tuning more of the network, and video models that keep the scale consistent across frames [40].

F. Conclusion

We asked whether fine-tuning a metric monocular depth model with a phone’s own LiDAR makes it more accurate than its pretrained self against a laser scanner. For objective (1) it did, on the rooms tested and by the criterion of the Introduction: FT-LiDAR-24 lowered the Chamfer distance from 54.3 to 15.7 cm in all six test rooms and AbsRel from 0.410 to 0.130 in 19 of 20 further rooms, with no calibration or sensor at run time and with weights learned from LiDAR alone, although within-1.25 $\times$ improved in only five of the six test rooms. For objective (2), laser labels were not measurably better than LiDAR labels, nor 24 rooms better than 10, within the limited power of six rooms, and the true focal length did not make Depth Pro competitive. For objective (3), the model lies between a relative model calibrated once per room and one aligned on sparse points in every frame, serves none of the three use cases, and is no substitute for iPad LiDAR (3.0 cm). For objective (4), its remaining error is a scale that varies with the view rather than drifting with time, whose bias fine-tuning removed but whose spread it did not. Per-frame metric

points from the device’s tracker are the most promising next step; their combination with the fine-tuned model remains to be tested.

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